Shen-Ning Tung

Curriculum Vitae

Department of Mathematics, National Tsing Hua University
Room 515, General Building III
101, Section 2, Kuang-Fu Road, Hsinchu 300, Taiwan
tung@math.nthu.edu.tw • sntung41406.github.io

Research Interests

Current focus: Mathematical and digital finance; automated market makers and prediction markets; decentralized finance; statistical modelling and machine learning.

Methods: Stochastic modelling and control; game theory; representation learning.

Earlier research: Algebraic number theory and p-adic Langlands.

Academic Positions

2022–present **Assistant Professor**
Department of Mathematics, National Tsing Hua University
One-year research leave in 2025

2019–2022 **Postdoctoral Fellow**
Department of Mathematics, University of British Columbia
Advisors: Rachel Ollivier and Sujatha Ramdorai

Education

2015–2019 **Ph.D. in Mathematics**, Universität Duisburg-Essen
Advisor: Vytautas Paškūnas
Thesis: *On the automorphy of 2-dimensional potentially semi-stable deformation rings of* $G_{\mathbb{Q}_p}$

2013–2015 **M.Sc. in Mathematics**, University of British Columbia
Advisor: Sujatha Ramdorai
Thesis: *Fontaine rings and p-adic L-functions*

2008–2012 **B.S. in Mathematics**, National Taiwan University

Grants and Funding

2025–2028 **NSTC Research Fellowship** – NTD 2,208,000
Mathematical Foundation of Automated Market Makers

2022–2025 **NSTC Research Fellowship** – NTD 2,116,000
Patching and the Completed Homology of Locally Symmetric Spaces

Publications

Peer-Reviewed Articles

1. **S.-N. Tung** and T.-H. Wang. A mathematical framework for modelling CLMM dynamics in continuous time. *Digital Finance*, 8, Article 39 (2026). doi:10.1007/s42521-026-00198-z
2. **S.-N. Tung**, C. Y. Lee, and T.-H. Wang. Growth rate of liquidity provider's wealth in G3Ms. *Applied Mathematical Finance*, 32(6), 379–422 (2025). doi:10.1080/1350486X.2026.2662662
3. A. C. Silva, **S.-N. Tung**, and W.-R. Chen. Stylized facts in Web3. *Frontiers of Mathematical Finance*, 3(4), 572–609 (2024). doi:10.3934/fmf.2024021
4. J. Najnudel, **S.-N. Tung**, K. Yamazaki, and J.-Y. Yen. An arbitrage driven price dynamics of Automated Market Makers in the presence of fees. *Frontiers of Mathematical Finance*, 3(4), 560–571 (2024). doi:10.3934/fmf.2024018

5. **S.-N. Tung**. On the automorphy of 2-dimensional potentially semistable deformation rings of $G_{\mathbb{Q}_p}$. *Algebra & Number Theory*, 15(9), 2173–2194 (2021). doi:10.2140/ant.2021.15.2173
6. **S.-N. Tung**. On the modularity of 2-adic potentially semi-stable deformation rings. *Mathematische Zeitschrift*, 298, 107–159 (2021). doi:10.1007/s00209-020-02588-4
7. V. Paškūnas and **S.-N. Tung**. Finiteness properties of the category of mod p representations of $\mathrm{GL}_2(\mathbb{Q}_p)$. *Forum of Mathematics, Sigma*, 9, e80 (2021). doi:10.1017/fms.2021.72

Preprints

1. C. Y. Lee, J. Risk, **S.-N. Tung**, and T.-H. Wang. Dynamics of the liquidity profile in concentrated liquidity provision: An SPIDE approach. *arXiv:2610.30539* (2026). [arXiv]
2. M.-R. Guan and **S.-N. Tung**. Pre-game paired-comparison modeling of professional League of Legends map outcomes. *arXiv:2609.08060* (2026). [arXiv]
3. J. Risk, **S.-N. Tung**, and T.-H. Wang. Pricing and hedging for liquidity provision in Constant Function Market Making. *arXiv:2603.01344* (2026). [arXiv]
4. J. Risk, **S.-N. Tung**, and T.-H. Wang. Dynamics of Liquidity Surfaces in Uniswap v3. *arXiv:2509.05013* (2025). [arXiv]

Teaching Experience

National Tsing Hua University (2022–present)

- MATH 153: Introduction to Mathematics (Fall 2026; current)
- MATH 582: Quantitative Finance II (Fall 2026; current)
- MATH 594: Numerical Computation with JAX (Spring 2026)
- MATH 570: Quantitative Finance I (Spring 2026)
- MATH 597: Mathematical Foundations of Data Science (Fall 2024)
- MATH 570: Financial Mathematics II (Spring 2024)
- MATH 570: Financial Mathematics I (Fall 2023)
- MATH 104: Calculus II (Spring 2023)
- MATH 103: Calculus I (Fall 2022)

University of British Columbia (2019–2022)

- MATH 317: Calculus IV (Vector Calculus) (Spring 2022)
- MATH 220: Mathematical Proof (Winter 2020, 2021)
- MATH 253: Multivariable Calculus (Summer 2020)
- MATH 105: Integral Calculus with Applications to Commerce and Social Sciences (Spring 2020)
- MATH 104: Differential Calculus with Applications to Commerce and Social Sciences (Winter 2019)

Student Supervision

Current Students

2026–present	Sheng-Gang Huang and Chen-Yu Wen Undergraduate research mentees, supported by an NCTS AI grant *PolyWorld: Latent World Models for Autonomous Market Making on Prediction Markets*
2026–present	Chih-Yao Wu (Master's student)

Past Students

2025–2026	Hai-Ching Shih (Undergraduate Research Program)
2023–2024	De-Jin Huang, Chen-Chung Yang (Undergraduate Research Program)
2022–2024	Wei-Ru Chen (M.Sc., co-supervised with Yuki Chino)

Service and Professional Activities

2026	Course Committee Member, Department of Mathematics, National Tsing Hua University
2024	Mentor, Undergraduate Summer Research Program (USRP), National Center for Theoretical Sciences
2023–2025	Course Committee Member, Department of Mathematics, National Tsing Hua University
2022	Organizing Committee Member, Annual Meeting, Taiwanese Mathematical Society

Research Visits

Jul–Aug 2026	University of Calgary, Canada (Host: Jinniao Qiu)
Jul–Aug 2025	University of Calgary, Canada (Host: Jinniao Qiu)
Feb–Apr 2025	University of Calgary, Canada (Host: Jinniao Qiu)
Aug 2024	University of British Columbia, Canada (Host: Sujatha Ramdorai)
Dec 2020–Feb 2021	Academia Sinica, Taiwan (Host: Ming-Lun Hsieh)
May–Jun 2020	Academia Sinica, Taiwan (Host: Ming-Lun Hsieh)
Jul 2019	Academia Sinica, Taiwan (Host: Ming-Lun Hsieh)
Jul 2019	Morningside Center of Mathematics, China (Host: Yongquan Hu)
May 2018	University of Cambridge, UK (Host: Jack Thorne)

Selected Invited Talks

Mar 2026	Mathematics Colloquium, National Chengchi University
Mar 2025	Mathematics Colloquium, National Sun Yat-sen University
Feb 2025	Mathematical Finance Seminar, University of Calgary
May 2024	TWSIAM Annual Meeting, National Chung Hsing University
Feb 2020	Number Theory Seminar, University of Toronto
Jul 2019	Number Theory Seminar, Academia Sinica
Jul 2019	Number Theory Seminar, Morningside Center of Mathematics
Jul 2018	Number Theory Seminar, University of Bonn
May 2018	Number Theory Seminar, University of Cambridge
Mar 2018	Number Theory Seminar, University of British Columbia

References

Professor Tai-Ho Wang
Department of Mathematics
Baruch College, CUNY
tai-ho.wang@baruch.cuny.edu
Tel: +1 (646) 312-4130

Professor Jinniao Qiu
Department of Mathematics and Statistics
University of Calgary
jinniao.qiu@ucalgary.ca

Professor Ju-Yi Yen *(for teaching)*
Department of Mathematical Sciences
University of Cincinnati
ju-yi.yen@uc.edu
Tel: +1 (513) 556-4214

Dr. A. Christian Silva
Industry collaborator
IdataFactory
csilva@idatafactory.com